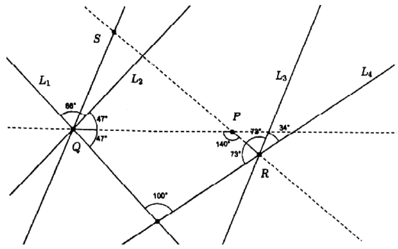
AMTI (NMTC) - 2015

GAUSS CONTEST - FINAL - PRIMARY LEVEL

1. A three digit number is divisible by 7 and 8.
   a) How many such numbers are there ?
   b) List out all the numbers.
   c) Find the two numbers whose digit sum is maximum and minimum.
   d) For how many numbers the digit sum is odd ?
2. a is the least number which on being divided by 5, 6, 8, 9 and 12 leaves in each case a remainder 1, but when divided by 13 leaves no remainder. b is the greatest 4–digit number which when divided by 12, 18, 21 and 28 leaves a remainder 3 in each case. Find the value of (b – a).
3. L_1, L_2, L_3, L_4 are straight lines such that L_1, L_2 intersect at Q and L_3, L_4 intersect at R in the same plane as in the diagram. The two dotted lines are the bisectors of the respective angles exterior to 86^0 and 34^0 and they meet at P. If L_1 and L_4 make an angle 100^0, find the measure of $\angle QPR$. What is the angle between the lines L_2 and L_3 ?



4. The two digit number 27 is 3 times the sum of the digits, since (2 + 7) x 3 = 27. Find all two digit numbers each of which is 7 times the sum of its digits.
5. There are a ten digit number a b c d e f g h i j with a = 1 and all the other digits are equal to either 0 or 1. It has the property that a + c + e + g + i = b + d + f + h + j. How many such 10-digit numbers are there ?
6. All the natural numbers from 1 to 12 are written on 6 separate pieces of paper, two numbers on each piece. The sums of the numbers on these six pieces are respectively 4, 6, 13, 14, 20 and 21. Find the pairs of the integers written on each piece of paper.
